Xu *et al. BMC Oral Health* (2025) 25:1190
https://doi.org/10.1186/s12903-025-06571-5

BMC Oral Health

RESEARCH Open Access

# Correlation between latent categories of body image and sleep quality in adolescent orthodontic patients

Chen Xu[1†], Ying Zhang[1†], Bin Xiong[2†], Hualun Wang[1] and Mengyao Liang[3*]

## Abstract

**Objective** This study aimed to identify potential categories of negative body image among adolescent orthodontic patients and further examine the relationship between these categories and sleep quality.To provide evidence for clinical psychological intervention.

**Methods** A total of 322 adolescent orthodontic patients were randomly selected from six stomatological hospitals in Jiangsu and Zhejiang provinces, China, between June 2024 and February 2025. A field survey was conducted using a general information questionnaire, the Sociocultural Attitudes Towards Appearance Questionnaire-4 (SATAQ-4R), the Pittsburgh Sleep Quality Index (PSQI), and the Body Image Scale (BISS). Data were analyzed using latent profile analysis, $\chi^2$ tests, and logistic regression.

**Results** Of the 322 surveys distributed, 311 valid responses were collected, yielding an effective response rate of 96.58%. The average body image score was 29.36 ± 6.13. Latent profile analysis identified three categories of body image: self-acceptance (11.90%, 37 cases), self-negative (60.77%, 189 cases), and self-critical (27.33%, 85 cases). $\chi^2$ tests revealed significant differences related to gender, being an only child, education level, BMI, and SATAQ-4R scores ($P < 0.05$). Logistic regression analysis indicated that female gender, self-negative, and self-critical body image types were significant risk factors for poor sleep quality in this population ($P < 0.05$).

**Conclusions** Adolescent orthodontic patients exhibit heterogeneous body image perceptions, with negative body image serving as a risk factor for poor sleep quality. Clinical attention should be paid to female and adolescent orthodontic patients with higher appearance anxiety or peer pressure. Early psychological assessment and combined orthodontic-psychological intervention should be used to help improve their body image cognition and sleep quality..

**Keywords** Adolescent, Orthodontics, Body image, Quality of sleep

[†]Chen Xu, Ying Zhang and Bin Xiong contributed equally to this work.

*Correspondence:
Mengyao Liang
2216314456@qq.com

[1]Department of Prosthetics, Affiliated Nantong Stomatological Hospital of Nantong University, Nantong 226000, China
[2]Department of Orthodontics, Affiliated Nantong Stomatological Hospital of Nantong University, Nantong 226000, China
[3]Department of Nursing, The Sixth People's Hospital of Nantong, Nantong 226001, Jiangsu, China

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

## Introduction

Orthodontic treatment in adolescents is a widely used approach to correct abnormal dental and maxillofacial development, aiming to enhance occlusal function, facial aesthetics, and mental health by aligning teeth and improving jaw relationships [1]. The period between 12 and 18 years is considered optimal for orthodontic intervention, as the maxillofacial bones are still developing, offering greater plasticity and more pronounced treatment effects [2]. Common orthodontic issues, such as overcrowded or spaced teeth, misalignment, and jaw dysplasia, not only affect oral function but also have significant psychological and social implications for adolescents [3].

Body image refers to an individual's subjective perception and evaluation of their body, encompassing aspects such as appearance, body function, health, and gender identity [4]. This multi-dimensional construct is shaped by a combination of physiological, psychological, social, and cultural factors [5, 6]. For adolescent orthodontic patients, abnormal dental or maxillofacial development can lead to negative body image perceptions, as features like misaligned teeth or facial asymmetry may induce feelings of inferiority and social withdrawal [7, 8]. Research suggests that negative body image not only impacts mental well-being but may also decrease treatment adherence, indirectly influencing treatment outcomes [9]. Furthermore, the relationship between body image and orthodontic treatment is bidirectional: while treatment can improve body image by enhancing the appearance of the teeth and face, a patient's initial body image perception can also affect their attitudes and expectations towards treatment. Adolescents with negative body image may have heightened expectations for treatment, which, if unmet, can increase psychological stress and dissatisfaction [10, 11]. Thus, monitoring psychological well-being and body image changes throughout orthodontic treatment is crucial.

In recent years, adolescent sleep quality has garnered increasing attention. Numerous studies indicate that adolescents commonly experience issues such as insufficient sleep, poor sleep quality, insomnia, and disrupted sleep patterns [12]. These sleep disturbances not only impair physical health—such as weakened immunity and stunted growth—but also negatively affect mental well-being, contributing to issues like emotional dysregulation and cognitive decline [13]. Notably, there appears to be a complex interplay between sleep quality and body image, where negative body image may indirectly compromise sleep by triggering emotional problems like anxiety and depression. For instance, adolescents dissatisfied with their body image may ruminate on their appearance at night, leading to difficulties falling asleep or disrupted sleep [14]. Conversely, poor sleep quality can exacerbate body image concerns. Research suggests that sleep deprivation may distort appearance perception, making individuals more likely to focus on perceived "flaws," thereby reinforcing a harmful cycle [12].

Although studies have addressed body image in adolescents, there are three key gaps in research addressing orthodontic patient populations: First of all, the time dimension is limited. The existing literature focuses on the middle and late stages of orthodontic treatment (such as 6 months after appliance wearing) [15], and lacks systematic evaluation of the baseline psychological state before treatment, which results in the inability to distinguish the effect of treatment itself from the original psychological characteristics of patients. Secondly, there are methodological limitations. Traditional variable center analysis (such as regression model) [16] fails to reveal the underlying subtype structure of body image, and there may be qualitative differences in psychological responses of orthodontic patients. Finally, there is a lack of group specificity. Current sleep research mainly focuses on ordinary adolescents [17], ignoring the unique psychological mechanism of orthodontic patients. Therefore, this study aimed to analyze the latent category characteristics of body image in adolescent orthodontic patients and to quantify the differential effects of different body image categories on sleep quality, thereby providing an evidence-based basis for the clinical establishment of a psychological risk stratification intervention system.

The study focused on adolescent patients who had not yet undergone orthodontic treatment, a methodological choice with significant implications. First, patients in the pre-treatment stage provide a clearer reflection of their baseline body image and sleep quality, free from the changes and disruptions caused by orthodontic devices. Second, this approach helps eliminate potential biases related to ongoing treatment, ensuring the scientific rigor and reliability of the findings.

## Methods

### Sample size calculation

The sample size was calculated based on a cross-sectional approach ($n = \frac{Z_\alpha{}^2 * P * (1-P)}{\delta^2}$) with a two-sided significance level of $\alpha = 0.05$. According to existing literature, the estimated prevalence of poor sleep quality in adolescents was 40%, with an allowable margin of error of 0.06. This calculation resulted in a required sample size of 257. Accounting for a 20% rate of invalid questionnaires, at least 309 participants were needed.

### Participants

Six stomatological hospitals in eastern China (Jiangsu Province and Zhejiang Province), covering the urban-rural fringe (2 hospitals) and urban centers (4 hospitals),

were selected for the survey. From June 2024 to February 2025, multi-stage random sampling was conducted to continuously include eligible patients to avoid seasonal treatment bias. Adolescents and their guardians were informed about the study's content, objectives, and confidentiality procedures prior to participation, with written informed consent obtained. Inclusion criteria were: (1) aged between 12 and 18 years; (2) patients scheduled for orthodontic treatment but not yet using orthodontic appliances; (3) ability to understand, cooperate with, and independently complete the questionnaire. Exclusion criteria included: (1) those undergoing initial orthodontic treatment; (2) Patients with severe skeletal Class II malocclusion (ANB > 8°) and confirmed OSA (AHI ≥ 5 events/hour) (3) Patients with respiratory diseases. (4) Patients with definite temporomandibular disorders (TMD). (5) Diseases known to be strongly associated with mouth breathing (e.g., allergic rhinitis, adenoid hypertrophy). In collaboration with hospital staff, the investigators distributed the paper-based questionnaires on-site, given that some adolescents lacked access to mobile phones. A total of 322 questionnaires were distributed, with 311 valid responses received, resulting in an effective recovery rate of 96.58%. The average age of the participants was 14.36 ± 2.26 years. There were 160 males (51.45%) and 151 females (48.55%). The study was approved by the Ethics Committee of Nantong Stomatological Hospital (Approval No. PJ-2024-022-01), and written informed consent was obtained from all guardians.

## Research tools

### *Self-made general information questionnaire*

Factors such as gender, educational background, being an only child, place of residence, regular exercise, and BMI were included as variables in the study.

### *Body image States scale (BISS)*

The BISS, developed by Cash et al. and revised by Yang Xiaoying et al. [18, 19], was used to assess body satisfaction. This 6-item scale utilizes a 9-point rating system, where 1 represents "very dissatisfied" and 9 represents "extremely satisfied." Higher scores reflect greater body satisfaction. The scale's internal consistency in this study was high, with a coefficient of 0.88.

### *Physical appearance social pressure scale*

The Sociocultural Attitudes Towards Appearance Questionnaire-4 (SATAQ-4R), developed by Schaefer et al. [20], was employed to measure external social pressures related to appearance. This 16-item scale is scored on a 5-point Likert scale, with 1 indicating "strongly disagree" and 5 indicating "strongly agree." Higher scores suggest greater perceived pressure from the media, family, peers, and significant others. The scale demonstrated good reliability within the Chinese population, with a Cronbach's α coefficient of 0.91 in this study.

### *Pittsburgh sleep quality index (PSQI)*

PSQI is an 18-item self-report questionnaire designed to assess sleep quality over the past month. It includes seven dimensions that comprehensively reflect the sleep status of the participant. The total PSQI score is the sum of the scores across these dimensions, with higher scores indicating poorer sleep quality. The scale was developed by Buysse et al. [21] and later translated and revised into Chinese by Liu Xianchen et al. [22]. PSQI scores range from 0 to 21, with higher scores correlating with decreasing sleep quality. A score of ≥ 11 indicates poor sleep quality.

## Quality control

Prior to the implementation of the policy, a pre-test of 20 questionnaires was conducted to ensure their accuracy, completeness, comprehensibility, and operability. All investigators underwent training led by the project leader to standardize instructions and testing procedures. During the questionnaire distribution phase, uniformity was maintained to minimize variations between centers. In the data entry phase, double entry was employed to ensure data accuracy and consistency. Logical checks and outlier detection were also performed to further validate the data's reliability and integrity.

## Statistical analysis

Data analysis was performed using SPSS 24.0 and Mplus 8.0 software. LPA was initially applied to classify the negative body image scores in adolescents. Model fit was evaluated using several indices: Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), adjusted Bayesian Information Criterion (aBIC), information entropy, Lo-Mendell-Rubin Likelihood Ratio Test (LMRT), and Bootstrap Likelihood Ratio Test (BLRT). A better model fit is indicated by smaller AIC, BIC, and aBIC values, an entropy value greater than 0.7 (with a value closer to 1 indicating higher classification accuracy), and significant results ($P < 0.05$) in both the LMRT and BLRT tests [23]. For the difference analysis, chi-square tests were used to assess variations in sleep quality across demographic characteristics and latent body image categories. Multivariate logistic regression analysis was then conducted, using sleep quality as the dependent variable and the variables with $P < 0.05$ from the univariate analysis as independent variables. The results were expressed as odds ratios (OR) with 95% confidence intervals (95% CI). Statistical significance was set at $P < 0.05$ for all analyses.

**Table 1** Fit index for latent profile analysis of body image in adolescent orthodontic patients

| Type | AIC | BIC | aBIC | Entropy | $P_{BLRT}$ | $P_{LMRT}$ |
|---|---|---|---|---|---|---|
| 1 | 8563.66 | 8621.57 | 8498.77 | — | — | — |
| 2 | 8503.34 | 8427.55 | 8486.83 | 0.817 | < 0.001 | < 0.001 |
| 3 | 8443.23 | 8318.93. | 8376.34 | 0.864 | < 0.001 | < 0.001 |
| 4 | 8329.67 | 8265.43 | 8317.84 | 0.794 | < 0.001 | < 0.001 |
| 5 | 8326.78 | 8210.34 | 8225.12 | 0.812 | 0.129 | 0.025 |
| 6 | 8209.89 | 8116.21 | 8167.98 | 0.743 | < 0.001 | 0.346 |

"-": No data.AIC: Akaike Information Criterion; BIC: Bayesian Information Criterion; aBIC: adjusted Bayesian Information Criterion;BLRT: Bootstrap Likelihood Ratio Test; LMRT: Lo-Mendell-Rubin Test

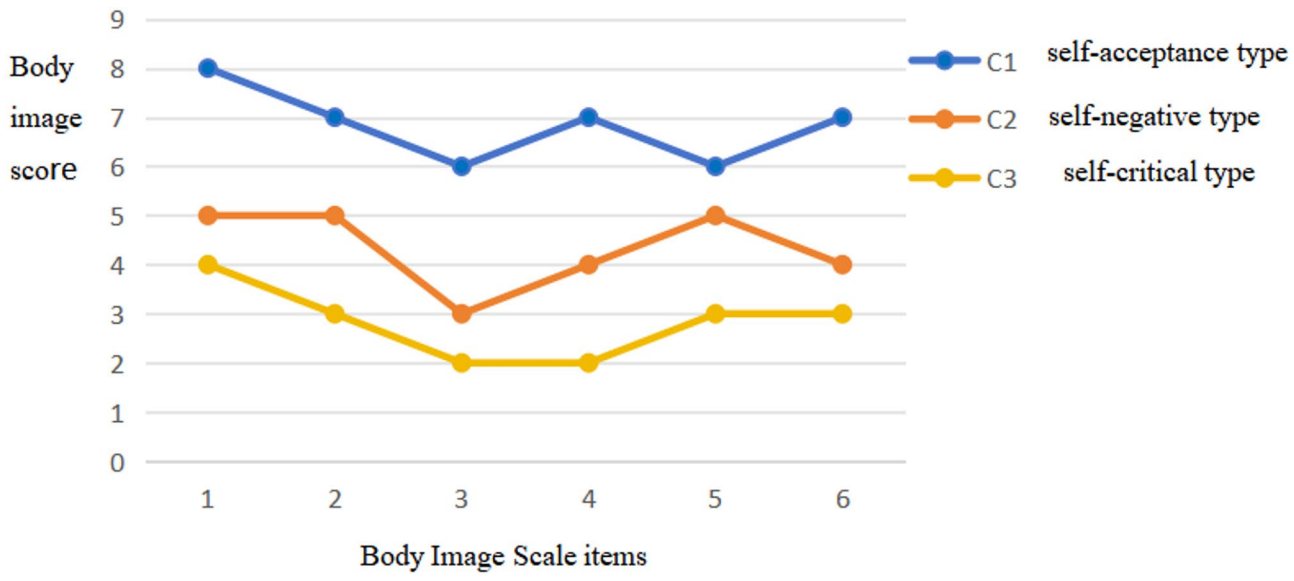


**Fig. 1** Characteristics distribution of three latent categories of body image in adolescent orthodontic patients

## Results

### Common method test

Nineteen factors had eigenvalues greater than 1, with the maximum variance explained by the factors being 22.34% (below the 40.00% threshold), indicating no common method bias in this study.

### Latent profile analysis of body image in adolescent orthodontic patients

The latent profile model based on body image scores was established, and the model fit results are presented in Table 1. As the number of model categories increased, the AIC, BIC, and aBIC values decreased. For the three-category model, the entropy value was 0.864, which was higher than for the other models. Additionally, both the BLRT and LMRT tests showed statistically significant differences ($P < 0.01$). Therefore, the three-category model best represents the body image characteristics of adolescent orthodontic patients (Table 1).

### Categories and naming of body image in adolescent orthodontic patients

The distribution of potential body image categories among adolescent orthodontic patients was as follows: Category 1 (self-acceptance type) consisted of patients who exhibited high scores across all items, indicating greater satisfaction with their body image and acceptance of their imperfections (11.9%, 37 cases). Category 2 (self-negative type) included patients who held negative attitudes toward their body image, believed their perceived defects negatively affected their overall image, and had relatively low scores across all items (60.77%, 189 cases). Category 3 (self-critical type) comprised patients who were overly critical of their body image, showing excessively low scores as they engaged in harsh self-evaluation and criticism (27.33%, 85 cases) (Fig. 1).

### Univariate analysis of influencing factors of potential categories of body image in adolescent orthodontic patients

Univariate analysis of the latent categories revealed statistically significant differences between groups in terms

**Table 2** Univariate analysis of body image in different types of adolescent orthodontic patients

| Variables | Classification | Total ($n=311$) | Self-acceptance type ($n=37$) | Self-negative type ($n=189$) | Self-critical type ($n=85$) | Value of statistics | *P* |
|---|---|---|---|---|---|---|---|
| Gender | Male | 160 | 24 | 107 | 29 | 15.360 | < 0.001 |
| | Female | 151 | 12 | 83 | 56 | | |
| Education | Junior high school and below | 178 | 13 | 107 | 58 | 10.831 | 0.004 |
| | High school | 133 | 23 | 83 | 27 | | |
| Whether they were the only child | No | 103 | 11 | 75 | 17 | 10.175 | 0.006 |
| | Yes | 208 | 25 | 115 | 68 | | |
| Exercise regularly | No | 99 | 14 | 61 | 24 | 1.339 | 0.512 |
| | Yes | 212 | 22 | 129 | 61 | | |
| BMI | < 18.5 | 58 | 14 | 37 | 7 | 33.097 | < 0.001 |
| | 18.5 ~ 24 | 133 | 20 | 68 | 45 | | |
| | 24 ~ 28 | 68 | 2 | 44 | 22 | | |
| | ≥ 28 | 52 | 0 | 41 | 11 | | |
| SATAQ- 4R* | | 41.22 ± 12.77 | 32.54 ± 6.54 | 39.81 ± 8.17 | 47.22 ± 6.21 | 14.765 | 0.002 |

Note: SATAQ- 4R: The Sociocultural Attitudes Towards Appearance Questionnaire- 4

**Table 3** Univariate analysis of sleep quality in different types of adolescent orthodontic patients

| Variables | Classification | Total ($n=311$) | Good sleep quality ($n=213$) | Poor sleep quality ($n=98$) | Value of statistics | *P* |
|---|---|---|---|---|---|---|
| Gender | Male | 160 | 119 | 41 | 5.852 | 0.016 |
| | Female | 151 | 94 | 58 | | |
| Education | Junior high school and below | 178 | 121 | 57 | 0.050 | 0.822 |
| | High school | 133 | 92 | 41 | | |
| Whether they were the only child | No | 103 | 70 | 143 | 0.020 | 0.888 |
| | Yes | 208 | 33 | 65 | | |
| Exercise regularly | No | 99 | 70 | 29 | 0.331 | 0.565 |
| | Yes | 212 | 143 | 69 | | |
| BMI | < 18.5 | 58 | 38 | 20 | 5.554 | 0.135 |
| | 18.5 ~ 24 | 133 | 98 | 35 | | |
| | 24 ~ 28 | 68 | 47 | 21 | | |
| | ≥ 28 | 52 | 30 | 22 | | |
| SATAQ- 4R | | 41.22 ± 12.77 | 39.97 ± 12.55 | 42.56 ± 10.72 | -1.873 | 0.081 |
| Body image | Self-acceptance type | 37 | 32 | 5 | 8.149 | 0.017 |
| | Self-negative type | 189 | 127 | 62 | | |
| | Self-critical type | 85 | 54 | 31 | | |

Note: SATAQ- 4R: The Sociocultural Attitudes Towards Appearance Questionnaire- 4

of gender, education, being an only child, BMI, and SATAQ-4R scores ($P<0.05$) (Table 2).

### Distribution of sleep quality in adolescent orthodontic patients with different variables

Among the 311 adolescents in this study, the prevalence of poor sleep quality was 37.62%. Significant differences in sleep quality were observed across gender, body image categories, and whether participants were the only child ($P<0.05$) (Table 3).

### Multivariate logistic regression analysis of sleep quality and body image in adolescent orthodontic patients

Using adolescent sleep quality as the dependent variable, factors with statistically significant differences from the univariate analysis were included in the multivariate logistic regression model. The results indicated that female adolescents (OR = 1.70, 95% CI: 1.032–2.803, $P<0.05$) and those with a self-negative body image (OR = 3.969, 95% CI: 1.344–11.714, $P<0.05$) had a significantly higher risk of poor sleep quality compared to male adolescents and those with a self-satisfactory body image (Table 4).

**Table 4** Multivariate logistic regression analysis of sleep quality in adolescent orthodontic patients

| The project | B-value | Standard error | Walds $\chi^2$ | *P* | OR | 95%CI |
|---|---|---|---|---|---|---|
| Constant | -2.282 | 0.543 | 17.638 | 0.001 | 0.102 | |
| Gender | | | | | | |
| Female | 0.531 | 0.255 | 4.348 | 0.037 | 1.701 | 1.032 ~ 2.803 |
| Body image categories | | | | | | |
| Self-negative type | 1.378 | 0.552 | 6.229 | 0.013 | 3.969 | 1.344 ~ 11.714 |
| Self-critical type | 1.524 | 0.576 | 6.999 | 0.008 | 4.593 | 1.485 ~ 14.208 |

## Discussion

The results of this study revealed that the body image of adolescent orthodontic patients was generally low, with an average body image score of 29.36 ± 6.13 (out of a total score of 54). This finding aligns with previous domestic research [24]. Several factors may contribute to this phenomenon. First, from a developmental psychology perspective, adolescence is a critical stage of physical and psychological growth, during which the imbalance between physical changes and psychological adjustment often leads to body image distress [25]. Second, orthodontic patients typically present with pre-existing issues such as misaligned teeth and facial asymmetry, which can heighten their awareness and sensitivity to their appearance [26, 27]. Additionally, social and cultural influences play a significant role. The widespread societal standards of beauty and the pervasive dissemination of aesthetic ideals can lead adolescent orthodontic patients to develop unrealistic expectations regarding their appearance [28].

In this study, latent profile analysis of body image in adolescent orthodontic patients revealed three distinct types: self-acceptance, self-negative, and self-criticism, comprising 11.9%, 60.77%, and 27.33% of the sample, respectively. This indicates significant heterogeneity within the group. However, there is a lack of research—both domestically and internationally—focusing on this specific population, and no prior latent profile analysis has been conducted in this context. The majority (60.77%) of participants exhibited a moderate body image, with low scores in "satisfaction with overall appearance" and "self-attractiveness." These patients were in a negative state regarding their self-image, marked by a lack of self-confidence and self-esteem. This could be attributed to the social, psychological, and cultural context surrounding young orthodontic patients. According to Sociocultural Theory, the idealized images promoted by mass media and society can foster unrealistic aesthetic expectations [29]. In the age of social media, adolescents are particularly vulnerable to the "appearance-first" culture, which often leads to negative body image [28]. Additionally, 27.33% of patients displayed a highly critical body image, scoring low in items such as "Self-image compared with the average person," indicating they viewed themselves as inferior to their peers. According to Trauma Theory, negative life experiences such as school bullying and appearance-related discrimination can have long-lasting effects on adolescents' body image [30]. These negative experiences, including bullying, ridicule, and social exclusion, can contribute to dissatisfaction, anxiety, and a critical self-evaluation. If adolescent orthodontic patients perceive their appearance as inadequate or not meeting societal aesthetic standards, they may develop feelings of inferiority, helplessness, and depression, ultimately impacting their self-esteem and self-confidence [31]. Persisting negative self-evaluation can have enduring effects on their body image, potentially influencing their daily life and academic performance. Thus, orthodontic care should include strategies to help adolescent patients establish a healthy body image and self-evaluation, mitigating excessive focus on appearance. Additionally, supporting them in managing social pressures regarding their looks and enhancing their self-esteem and self-confidence is essential.

The results of this study indicate that being female, having a self-negative body image, and having a self-critical body image are independent risk factors for poor sleep quality in adolescent orthodontic patients ($p < 0.05$). From a gender perspective, several factors may contribute to the poorer sleep quality observed in female patients. First, from a developmental psychology standpoint, adolescence is a critical phase for physiological changes in females, with the emergence of secondary sexual characteristics often leading to body image distress [32]. Research has shown that adolescent females tend to experience greater dissatisfaction with their weight and body shape compared to males [33], and such negative perceptions may disrupt sleep quality through neuroendocrine mechanisms [34]. Furthermore, social and cultural pressures also play a significant role. Females often face greater social comparison and the influence of social media, which can contribute to emotional regulation difficulties and, consequently, impaired sleep quality [35]. Regarding body image categories, both the self-negative and self-critical groups exhibited significantly poorer sleep quality than the self-acceptance group ($p < 0.05$). This can be explained through several mechanisms: First, according to cognitive behavioral theory, negative body image is linked to increased rumination, where persistent negative thoughts activate the sympathetic nervous

system, disrupting the sleep-wake cycle [36]. Additionally, Social Comparison Theory posits that excessive upward social comparison lowers self-esteem and raises anxiety, which, in turn, may impair sleep quality through the hypothalamic-pituitary-adrenal (HPA) axis [37].

The results of this study showed that there was a significant correlation between psychological status and sleep quality in adolescent orthodontic patients, and systematic intervention strategies were needed to integrate psychological support and orthodontic treatment. Based on the results of this study, the following clinical practice recommendations are proposed: in the initial diagnosis stage, it is recommended to integrate standardized psychological assessment tools, such as BISS and PSQI, into the routine examination process. High-risk patient profiles were established by identifying the body image subtypes of patients through LPA. During the follow-up period, changes in sleep quality were tracked every 3 months, and if the PSQI total score was significantly worse than the baseline level, a secondary assessment with the participation of a psychologist was initiated. For self-acceptance patients, 3D digital models are used to dynamically display the treatment progress (such as the improvement of maxillary arch width and the increase of occlusal contact point), combined with the comparison map of stage occlusion relationship, so as to strengthen the positive cognition of patients on functional improvement [38]. For self-negative patients, cognitive behavioral intervention techniques can be integrated into chair-side operations, such as simultaneous interpretation of functional benefits when adjusting appliances (e.g., alignment can reduce the frequency of food impaction) [39].

For self-critical patients, a multidisciplinary collaboration mechanism is established to identify negative experiences related to appearance (such as being derated for tooth protrusion) through traumatic event diary, and mindfulness regulation strategies (deep breathing combined with body scanning meditation) are taught to block rumination at night [40]. It is suggested to add the core module of "psychological support" in orthodontic continuing education, focus on training empathy communication skills and use of non-critical language to build trust between doctors and patients. The above integrated strategies can provide physical and mental synergy intervention for adolescent orthodontic patients, which has important clinical significance for improving treatment compliance and long-term prognosis.

This study has several limitations. First, while employing a multi-center design across Jiangsu-Zhejiang provinces, the regional socioeconomic homogeneity of participants may constrain generalizability to populations with divergent orthodontic accessibility or body image perceptions. Second, we did not systematically assess key confounding variables including psychosocial factors (anxiety/depression comorbidity, academic stress, family dysfunction) and physiological factors (caffeine intake, circadian rhythm disorders), limiting causal interpretation.

Third, measurement constraints warrant consideration: (a) Exclusive reliance on adolescent self-reported data introduces validity concerns—despite using validated instruments, potential question misinterpretation, emotional volatility, and social desirability bias persist; (b) All instruments were administered consecutively within a single 20-minute pre-treatment session, risking fatigue-induced inaccuracies, particularly for later questionnaires; (c) The PSQI's retrospective 1-month timeframe, though clinically practical, may inadequately capture acute sleep disturbances during orthodontic treatment anticipation phases; (d) Existing scales lack orthodontic-specific metrics (e.g., tooth protrusion/smile symmetry concerns), potentially overlooking subtle psychological reactions. Fourth, unmeasured clinical parameters preclude nuanced analysis: Documentation gaps regarding appliance types and treatment progress prevent assessment of how therapeutic stages modulate psychology-sleep dynamics. Fifth, while focusing on direct body image-sleep correlations, we acknowledge that abnormal craniofacial morphology (e.g., mandibular retrusion) may exert indirect effects through pathways like upper airway dysfunction (OSA) or neurovegetative alterations [41, 42]. Such mechanisms may be potentiated by anatomical characteristics specific to Chinese populations [43, 44]. Finally, the absence of objective sleep measures (actigraphy) or multi-informant validation precludes data triangulation, weakening causal inferences. Future research should integrate multidisciplinary evaluations (cephalometry, polysomnography) to delineate these complex biomechanical-psychological interactions.

## Conclusion

This study examined the correlation between latent categories of body image and sleep quality in adolescent orthodontic patients. The findings indicated that body image levels were generally low, with three latent categories identified: self-acceptance, self-negative, and self-critical. Notably, most patients exhibited negative body image. Female patients and those with self-negative or self-critical body image profiles were independent risk factors for poor sleep quality. Potential clinical applications emerging from this research include: developing a rapid screening tool using LPA classification to identify high-risk patients early; establishing a phased psychological intervention pathway to optimize treatment compliance; and advancing orthodontic care's transition from a biomedical toward a biopsychosocial model.

**Author contributions**
ML and HW designed this study.CX and YZ built the study framework, BX and MLwrote the manuscript, all contributed to the article and approved the submitted version.

**Funding**
Supported by Nantong University Special Research Fund for Clinical Medicine (Grant No. 2024LY066).

**Data availability**
The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Ethics approval and consent to participate**
The Ethics Committee of Affiliated Nantong Stomatological Hospital of Nantong University provided approval for conducting this study (approval number: PJ 2024-022-01). All patients provided written informed consent. In addition to this, all the methods/procedures were performed in accordance with the relevant guidelines and regulations (Declaration of Helsinki). For all participants under the age of 16, written/verbal informed consent was obtained from their parents or legal guardians prior to participation.

**Consent for publication**
Not applicable.

**Informed consent**
All patients provided informed consent.

**Competing interests**
The authors declare no competing interests.

Received: 30 April 2025 / Accepted: 8 July 2025
Published online: 17 July 2025

**References**

1. Motoyoshi M, Matsuoka M, Shimizu N. Application of orthodontic Mini-Implants in adolescents. INT J ORAL MAX SURG. 2007;36:695–9. https://doi.org/10.1016/j.ijom.2007.03.009.
2. Twigge E, Roberts RM, Jamieson L, Dreyer CW, Sampson WJ. The Psycho-Social impact of malocclusions and treatment expectations of adolescent orthodontic patients. EUR J ORTHODONT. 2016;38:593–601. https://doi.org/10.1093/ejo/cjv093.
3. Gavic L, Budimir M, Tadin A. The association between Self-Esteem and aesthetic component of smile among adolescents. PROG ORTHOD. 2024;25:9. https://doi.org/10.1186/s40510-023-00508-w.
4. Johal A, Damanhuri SH, Colonio-Salazar F, Adult, Orthodontics. Motivations for treatment, choice, and impact of appliances: A qualitative study. AM J ORTHOD DENTOFAC. 2024;166:36–49. https://doi.org/10.1016/j.ajodo.2024.02.009.
5. Grogan S. Body image and health: contemporary perspectives. J HEALTH PSYCHOL. 2006;11:523–30. https://doi.org/10.1177/1359105306065013.
6. Saylan E, Soyyigit V. Body image among adolescents: what is its relationship with rejection sensitivity and Self-Efficacy? CLIN CHILD PSYCHOL P. 2024;29:479–92. https://doi.org/10.1177/13591045231188411.
7. Mahama S, Weichold K, Fehmer N, Mvungu EN, Natsuaki MN. Pubertal status and body image: an inquiry into experiences of adolescents in Ghana and Kenya. J RES ADOLESCENCE. 2024;34:257–71. https://doi.org/10.1111/jora.12953.
8. Paquette MC, Raine K. Sociocultural context of women's body image. SOC SCI MED. 2004;59:1047–58. https://doi.org/10.1016/j.socscimed.2003.12.016.
9. Roulias P, Vasoglou G, Angelopoulos G, Pandis N, Sifakakis I. Effect of aligners on patients' oral Health-Related quality of life and anxiety: A prospective pilot study. BMC PSYCHOL. 2024;12:346. https://doi.org/10.1186/s40359-024-01834-2.
10. Nie P, Teng F, Wang F, Shen G, Zhu M, Yu D. Correlation between esthetic expectations for orthognathic treatment and psychological characteristics among Chinese adult patients. AM J ORTHOD DENTOFAC. 2021;160:94–100. https://doi.org/10.1016/j.ajodo.2020.03.034.
11. Deng Xiao. Theoretical model of negative body image in adolescent orthodontic patients and its cognitive neural mechanism. Chongqing Medical University; 2015.
12. Xu Q, Du W, Lin F. What impact do personality traits have on Self-Perception of dental aesthetics?? HEAD FACE MED. 2023;19(11). https://doi.org/10.1186/s13005-023-00358-1.
13. Cavalcanti L, Lima RA, Silva C, Barros M, Soares FC. Constructs of poor sleep quality in adolescents: associated factors. CAD SAUDE PUBLICA. 2021;37:e207420. https://doi.org/10.1590/0102-311X00207420.
14. Ye Y, Wang H, Liao X, Yu C, Zhang W. Stressful life events, sleep quality and Non-Suicidal Self-Injury in Chinese adolescents: the moderating effect of sensation seeking. BEHAV SCI-BASEL. 2024;14286. https://doi.org/10.3390/bs14040286.
15. Rosenbaum DL, Gillen MM, Bloomer SA. The effects of sleep on body image: examining the roles of depression, perceived stress, and anxiety. J AM COLL HEALTH. 2024;72:3662–70. https://doi.org/10.1080/07448481.2023.2186153.
16. Imani MM, Jalali A, Dinmohammadi M, Nouri P. The effect of orthodontic intervention on mental health and body image. Open Access Maced J Med Sci. 2018;6:1132–7. https://doi.org/10.3889/oamjms.2018.243.
17. Black EB, Garratt M, Beccaria G, Mildred H, Kwan M. Body image as a predictor of nonsuicidal Self-Injury in women: A longitudinal study. COMPR PSYCHIAT. 2019;88:83–9. https://doi.org/10.1016/j.comppsych.2018.11.010.
18. Agostini A, Centofanti S, CLIN N AM. Normal sleep in children and adolescence. PSYCHIAT. 2024; 47:1–14. https://doi.org/10.1016/j.psc.2023.06.001
19. Cash TF, Fleming EC, Alindogan J, Steadman L, Whitehead A. Beyond body image as a trait: the development and validation of the body image States scale. EAT DISORD. 2002;10:103–13. https://doi.org/10.1080/10640260290081678.
20. Yang XY, Chen H, Gao X, Chen R. The role of social comparison in the influence of media on body image of female college students. Chin J Mental Health. 2010;24:51–4.
21. Schaefer LM, Harriger JA, Heinberg LJ, Soderberg T, Kevin TJ. Development and validation of the Sociocultural attitudes towards appearance Questionnaire-4-Revised (Sataq-4R). INT J EAT DISORDER. 2017;50:104–17. https://doi.org/10.1002/eat.22590.
22. Buysse DJ, Reynolds CR, Monk TH, Berman SR, Kupfer DJ. The Pittsburgh sleep quality index: A new instrument for psychiatric practice and research. PSYCHIAT RES. 1989;28:193–213. https://doi.org/10.1016/0165-1781(89)90047-4.
23. Liu XC, Tang MQ. HU L. Reliability and validity of the Pittsburgh sleep quality index. Chin J Psychiatry. 1996:103–7.
24. Yin K, Peng J, Zhang J. Applications of latent profile analysis in the field of organizational behavior. Progress Psychol Sci. 2020;28:1056–70. https://doi.org/10.3724/SP.J.1042.2020.01056.
25. Peng MH, Kang J, Deng X. The role of body image in adolescent orthodontic treatment. [J] Stomatology. 2017;35:489–93. https://doi.org/10.7518/hxkq.2017.05.008.
26. Carvalho GX, Nunes A, Moraes CL, Veiga G. Body image dissatisfaction and associated factors in adolescents. CIENC SAUDE COLETIVA. 2020;25:2769–82. https://doi.org/10.1590/1413-81232020257.27452018.
27. Deng X, Wang YJ, Deng F, Liu PL, Wu Y. Psychological Well-Being, dental esthetics, and psychosocial impacts in adolescent orthodontic patients: A prospective longitudinal study. AM J ORTHOD DENTOFAC. 2018;153:87–96. https://doi.org/10.1016/j.ajodo.2017.05.028.
28. Bahar AD, Sagi MS, Mohd ZF, Wan HW. Dental aesthetics and Self-Esteem of patients seeking orthodontic treatment. HEALTHCARE-BASEL. 2024;12:1576. https://doi.org/10.3390/healthcare12161576.
29. Demaria F, Pontillo M, Di Vincenzo C, Bellantoni D, Pretelli I, Vicari S, Body. Image, and digital technology in adolescence and contemporary youth culture. FRONT PSYCHOL. 2024;15:1445098. https://doi.org/10.3389/fpsyg.2024.1445098.
30. Vandenbosch L, Fardouly J, Tiggemann M. Social media and body image: recent trends and future directions. CURR OPIN PSYCHOL. 2022;45:101289. https://doi.org/10.1016/j.copsyc.2021.12.002.
31. Lian Q, Su Q, Li R, Elgar FJ, Liu Z, Zheng D. The association between chronic bullying victimization with weight status and body Self-Image: A Cross-National study in 39 countries. PEERJ. 2018;6:e4330. https://doi.org/10.7717/peerj.4330.
32. Jalali-Farahani S, Amiri P, Zarani F, Azizi F. A qualitative exploration of body image from the perspective of adolescents with a focus on psychological

aspects: findings from Iran. CHILD PSYCHIAT HUM D. 2023;54:202–12. https://doi.org/10.1007/s10578-021-01235-1.
33. Carlson JD. Body image among adolescent girls and boys: A longitudinal study. DEV PSYCHOL. 2004;40:823–35. https://doi.org/10.1037/0012-1649.40.5.823.
34. Hoffmann S, Cortes-Garcia L, Warschburger P. Weight/Shape and muscularity concerns and emotional problems in adolescent boys and girls: A Cross-Lagged panel analysis. J ADOLESCENCE. 2018;68:70–7. https://doi.org/10.1016/j.adolescence.2018.07.005.
35. Yu K, Xia L, Chen HH, et al. Association between sleep reactivity, Pre-Sleep arousal state, and neuroendocrine hormones in patients with chronic insomnia disorder. NAT SCI SLEEP. 2024;16:1907–19. https://doi.org/10.2147/NSS.S491040.
36. de Valle MK, Gallego-Garcia M, Williamson P, Wade TD. Social media, body image, and the question of causation: Meta-Analyses of experimental and longitudinal evidence. BODY IMAGE. 2021;39:276–92. https://doi.org/10.1016/j.bodyim.2021.10.001.
37. Akram U, Allen SF, Stevenson JC, Lazuras L, Ackroyd M, Chester J, Longden J, Peters C, Irvine KR. Homogeneity of cognitive and behavioural processes underlying the relationship between insomnia and body image disturbance. COGN PROCESS. 2021;22:701–9. https://doi.org/10.1007/s10339-021-01039-0.
38. Hu YT, Liu QQ, Ma ZF. Does upward social comparison on Sns inspire adolescent materialism?? Focusing on the role of Self-Esteem and mindfulness. J PSYCHOL. 2023;157:32–47. https://doi.org/10.1080/00223980.2022.2134277.
39. de Jong K, Douglas S, Wolpert M, et al. Using progress feedback to enhance treatment outcomes: A narrative review. ADM POLICY MENT HLTH. 2025;52:210–22. https://doi.org/10.1007/s10488-024-01381-3.
40. Chen Q, Liu Q, Wang W. Application effect of behavioral cognition combined with psychological intervention on orthodontic patients: A prospective, randomized, controlled trial. Medicine. 2024;103:e37131. https://doi.org/10.1097/MD.0000000000037131.
41. Yu X, Langberg JM, Becker SP. Dispositional mindfulness moderates the relation between brooding rumination and sleep problems in adolescents. SLEEP MED. 2022;90:131–4. https://doi.org/10.1016/j.sleep.2022.01.017.
42. Dadgar-Yeganeh A, Hatcher DC, Oberoi S. Association between degenerative temporomandibular joint disorders, vertical facial growth, and airway dimension. J WORLD FED ORTHOD. 2021;10:20–8. https://doi.org/10.1016/j.ejwf.2021.01.001.
43. Silva US, Cordero OA, Barzallo SV. Relationship between dental occlusion and brain activity: A narrative review. SAUDI DENT J. 2022;34:538–43. https://doi.org/10.1016/j.sdentj.2022.09.001.
44. O'Driscoll DM, Landry SA, Pham J, Young A, Sands SA, Hamilton GS, Edwards BA. The physiological phenotype of obstructive sleep apnea differs between Caucasian and Chinese patients. Sleep. 2019;42:zsz186. https://doi.org/10.1093/sleep/zsz186.

## Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.